

DTSA 5752

Practical Molecular Simulation Calculations

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Applied Scientist (66.7 / 100)	37.9 / 100
SPECIALIZATION Simulation Methods for Data Science		

38

CURRENT AVG SCORE
across 6 job roles

70

POTENTIAL SCORE
with recommended additions

+40

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Building on the previous courses, this course covers a number of practical topics for calculations in biotechnology, pharmaceutical design and materials modeling including calculating polymer properties, modeling the behavior of proteins, estimating drug binding affinities, again based around using common open-source modeling software.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	66.7	100.0	+33.3	HIGH
AI Engineer	41.1	71.3	+30.2	HIGH
Data Scientist	38.5	76.4	+38.0	HIGH
Machine Learning Engineer	36.9	76.7	+39.8	HIGH
ML Engineer	24.5	47.1	+22.7	MEDIUM
MLOps Engineer	19.8	45.7	+25.9	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

LLMs & Generative AI 1 of 6 roles

storage

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 MEDIUM Consider Business Communication content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

3 LOW Optional: LLMs & Generative AI — would broaden appeal to one additional role. Candidate terms: storage.

GAP KEYWORDS BY ROLE

Applied Scientist

66.7 → 100.0 (+33.3 pts)

biology

computational

research

storage

problems

models

technical

agents

AI Engineer

41.1 → 71.3 (+30.2 pts)

perception

data

models

learning

build

technical

systems

platform

Data Scientist

38.5 → 76.4 (+38.0 pts)

data

presentation

mitigation

insights

social

build

problems

approaches

Machine Learning Engineer

36.9 → 76.7 (+39.8 pts)

data

learning

tools

solutions

exploratory

business

development

science

ML Engineer

24.5 → 47.1 (+22.7 pts)

data

learning

machine

systems

models

build

problems

new

MLOps Engineer

19.8 → 45.7 (+25.9 pts)

data

systems

learning

machine

tools

models

development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.

